LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Ch 8 Test: Heredity

Chapter / Topic: Heredity

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Show crosses clearly with gametes and offspring wherever asked.

SECTION A

6 marks

Choose the correct option.

Q1. Tall pea plants with violet flowers were crossed with short pea plants with white flowers. All the progeny had violet flowers, but almost half of them were short. The genetic make-up of the tall parent is: **[1]**

(a) TTWW (b) TTww
(c) TtWW (d) TtWw

Q2. In human beings, the sex of the child is decided by: **[1]**

(a) the mother (b) the father
(c) both parents equally (d) the environment

Q3. In Mendel's monohybrid cross, the phenotypic ratio in the F_2 generation is: **[1]**

(a) 1 : 2 : 1 (b) 9 : 3 : 3 : 1
(c) 1 : 1 (d) 3 : 1

Q4. Genes are located on the: **[1]**

(a) chromosomes (b) ribosomes
(c) cell membrane (d) cytoplasm

Q5. When a pure tall (TT) pea plant is crossed with a pure short (tt) plant, the F_1 plants are: **[1]**

(a) all short (b) half tall and half short
(c) all tall (d) 3 tall : 1 short

Q6. **Assertion (A):** Human females have a perfect pair of sex chromosomes, both called X. **Reason (R):** All children inherit an X chromosome from their mother. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. What are dominant and recessive traits? Give one example of each from Mendel's experiments. **[2]**

OR

How do genes control traits? Explain with the example of plant height.

Q8. How is the sex of a child determined in human beings? **[2]**

SECTION C

6 marks

Q9. Mendel crossed pea plants with round, yellow seeds (RRYY) with plants having wrinkled, green seeds (rryy). **[3]**

(i) What were the F_1 seeds like? **[1]**

(ii) Write the F_2 phenotypic ratio. **[1]**

(iii) What does this cross show about the inheritance of the two traits? **[1]**

Q10. Show, with gametes and a checkerboard, the result of crossing two heterozygous tall pea plants (Tt × Tt). Give the phenotypic and genotypic ratios. **[3]**

OR

In a cross between pure tall (TT) and pure short (tt) pea plants, the short trait disappears in F_1 but reappears in F_2. Explain why.

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) Describe Mendel's experiment on plant height. Give the F_1 and F_2 results. **[3]**

(ii) Give two reasons why the garden pea was a good choice for Mendel's experiments. **[2]**

OR

(a) What is variation? Why is it important for a species? [2]
(b) Explain sex determination in humans with the help of a cross. [3]

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. A student crossed pea plants with violet flowers (VV) with plants having white flowers (vv). All F_1 plants had violet flowers. When the F_1 plants were self-pollinated, out of 400 F_2 plants, about 300 had violet flowers and 100 had white flowers. **[4]**

(i) Which flower colour is dominant? **[1]**

(ii) Write the genotype of the F_1 plants. **[1]**

(iii) Out of the 300 violet-flowered F_2 plants, how many are expected to be pure-breeding (VV)? Explain. **[2]**

~ All the best! ~